

EX. NO: 7

COLUMNAR TRANSPOSITION CIPHER

Aim:

To write a C program to implement the Columnar Transposition Cipher, a transposition technique that rearranges plaintext characters by reading a filled matrix column by column.

Algorithm:

1. Start the program.
2. Read the plaintext and the number of columns from the user.
3. Calculate the number of rows required to accommodate the plaintext.
4. Fill the matrix row-wise with the characters of the plaintext, padding any remaining cells with the letter 'X'.
5. Read the matrix column by column to generate the encrypted text.
6. Display the resulting encrypted text.
7. Stop the program.

Program:

```
#include <stdio.h>
#include <string.h>

int main() {
    char text[100];
    int cols;

    printf("Enter Plain Text: ");
    scanf("%s", text);
    printf("Enter Number of Columns: ");
    scanf("%d", &cols);

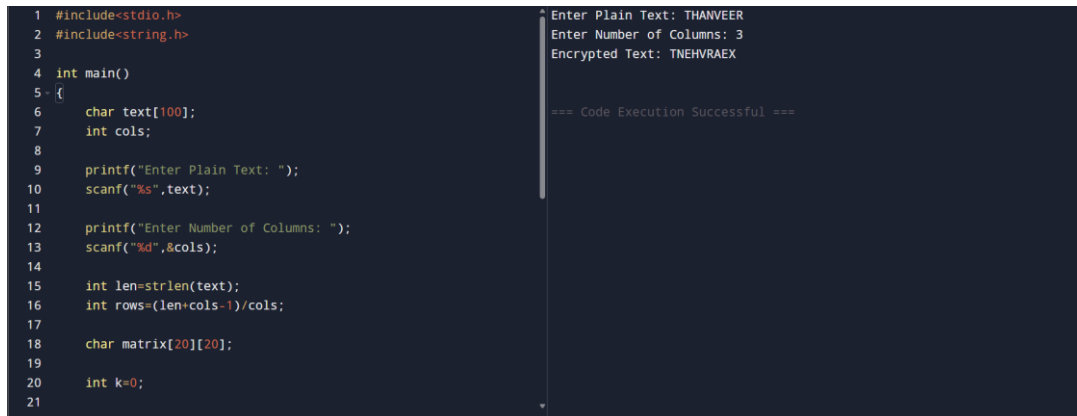
    int len = strlen(text);
    int rows = (len + cols - 1) / cols;
    char matrix[20][20];
    int k = 0;

    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            if (k < len)
                matrix[i][j] = text[k++];
            else
                matrix[i][j] = 'X';
        }
    }

    printf("Encrypted Text: ");
```

```
    for (int j = 0; j < cols; j++) {  
        for (int i = 0; i < rows; i++)  
            printf("%c", matrix[i][j]);  
    }  
    printf("\n");  
    return 0;  
}
```

Output:



```
1 #include<stdio.h>  
2 #include<string.h>  
3  
4 int main()  
5 {  
6     char text[100];  
7     int cols;  
8  
9     printf("Enter Plain Text: ");  
10    scanf("%s",text);  
11  
12    printf("Enter Number of Columns: ");  
13    scanf("%d",&cols);  
14  
15    int len=strlen(text);  
16    int rows=(len+cols-1)/cols;  
17  
18    char matrix[20][20];  
19  
20    int k=0;  
21
```

Enter Plain Text: THANVEER
Enter Number of Columns: 3
Encrypted Text: TNEHVRAEX

=== Code Execution Successful ===

Result:

Thus a C program to implement the Columnar Transposition Cipher was written, executed and the output was verified successfully.